CHANDLER PARK ACADEMY

Classroom Observation Guide

Grade 6 Mathematics — Day One: “What Is Math?”

Professor Xavier Honablue, M.Ed.

Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. This guide explains the structure of today's lesson, the Common Core and Michigan academic standards it addresses, and the educational research behind the instructional choices you will see in action. It is intended to give context to what may look like play with paper and shapes, but is in fact a deliberately designed introduction to mathematical thinking.

Lesson at a Glance (55 Minutes)

Time	Station	What's Happening
0-3 min	**Welcome**	Students settle in and meet Professor Xavier Honablue, M.Ed.
3-11 min	**Engage**	Introduction to the E3 Math Station System and the first-3-days roadmap.
11-15 min	**Engage**	Definition of math introduced: a numerical or graphical representation of an observation, or a simulation of an imagined observation.
15-30 min	**Engage**	Whole-class observation of a real dragonfly; students simulate it with geometric shape manipulatives along a quadded, diagonal graph-paper line; build a shape tally.
30-42 min	**Explore**	Working with classmates at the back board, students design and build their own imagined object on a second, self-quadded sheet, keeping the same tally system.
42-49 min	**Enrich**	Small-group time with Professor Xavier to learn how a shape tally can be used to name the simulated object.
49-52 min	**Turn It In**	Completed graph-paper sheets are

		submitted in the class collection box.
52-55 min	**Reflect**	Standards are named in student-friendly language and students reflect on what they accomplished.

The E3 Math Station System

Every lesson at Chandler Park Academy moves students through three stations, each with a distinct instructional purpose:

- **ENGAGE — Front Smart Board & Webcam:** Whole-class instruction at the front smart board and webcam. Professor Xavier models the day's task so every student begins with a shared understanding.
- **EXPLORE — Back Board with Friends:** Students work with classmates at the back board using dry-erase markers to try the task themselves before working independently.
- **ENRICH — Enrich Area:** Students meet with Professor Xavier in a small group or one-on-one for deeper learning, clarification, and an extension challenge.

Student Work Sample

Below is a representative sample of the Engage/Explore task, built with geometric shape manipulatives along a diagonal line. Notice how the student combined triangles, parallelograms, a trapezoid, and a hexagon to approximate a real, organic shape using only straight-edged pieces — a direct illustration of MP4 (Model with Mathematics) and MP6 (Attend to Precision) in action.

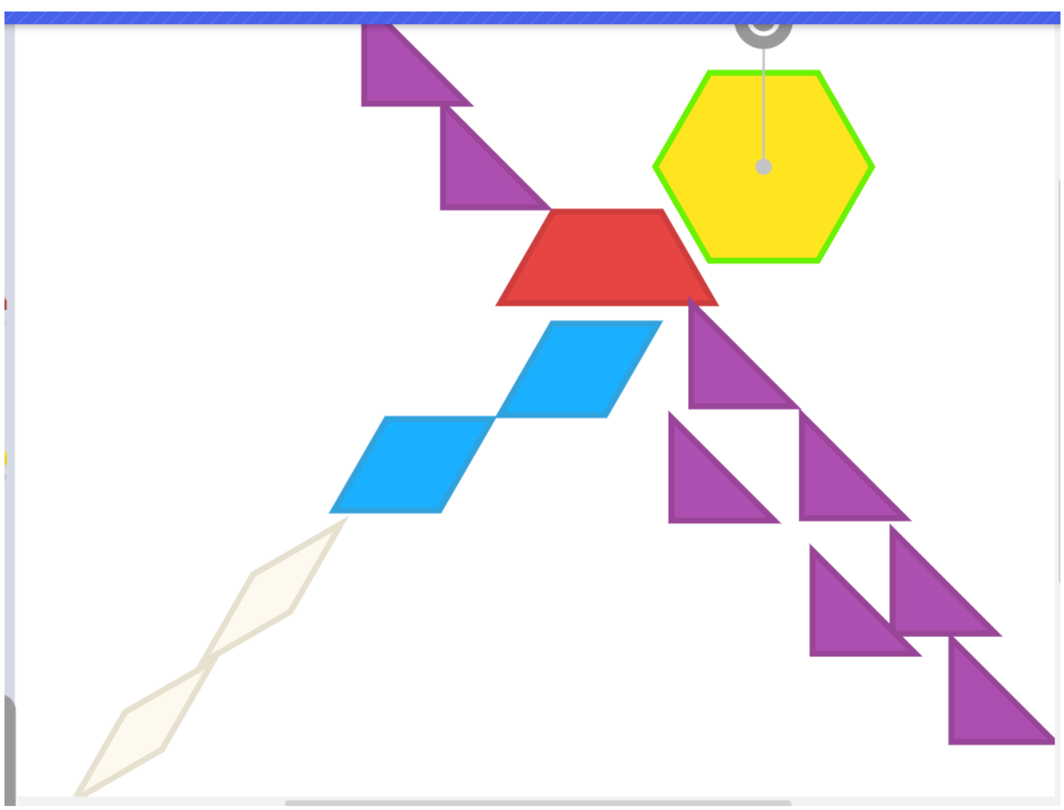

Figure 1. A student's shape-based simulation, built along the diagonal line described in the Engage station.

Common Core & Michigan Standards Addressed

Michigan's State Board of Education adopted the Common Core State Standards in 2010; the resulting Michigan K-12 Standards for Mathematics use the same codes and content as the national Common Core State Standards for Mathematics. Because this is the first lesson of the year, before any specific content strand begins, the primary standards addressed are the Standards for Mathematical Practice — the eight practices that describe what it looks like to “do” mathematics, and which anchor every content standard taught for the rest of the year.

MP4 — Model with Mathematics

Mathematically proficient students can apply what they know to represent a real situation using a diagram, drawing, or set of symbols. Today, students model a real, observed object (a dragonfly) using geometric shapes — the same core skill that, later in the year, will let them model area, patterns, and algebraic relationships.

MP5 — Use Appropriate Tools Strategically

Students consider which tools — graph paper, a ruler, geometric manipulatives — will help them build an accurate model, and use those tools with intention rather than by default.

MP6 — Attend to Precision

Students are asked to communicate precisely: constructing straight, exact lines and carefully counting and recording each shape they use. This habit of precision is explicitly named as a K-12 mathematical practice, not just a classroom-management expectation.

Content Standards Foreshadowed

While Day One is a practices-focused lesson, its structure previews content the class will formalize soon: 6.G.A.3 (drawing and constructing geometric figures), 6.G.A.1 (area of shapes built from simpler figures), and 6.SP.B.4 (organizing and displaying data), which the shape tally quietly introduces well before the class reaches a formal data unit.

Pedagogical & Cognitive Research Foundations

The instructional choices in today's lesson are not arbitrary. Each reflects a specific, evidence-based approach from mathematics education and cognitive science research.

Concrete-Representational-Abstract (CRA) Sequence

Students begin at the concrete level, handling physical geometric manipulatives, before moving to the representational level, drawing their simulated object on graph paper. This progression is grounded in CRA research (Witzel, Riccomini, & Schneider, 2008), which has repeatedly shown that sequencing instruction from concrete materials to pictorial representation, and only later to abstract symbols, improves conceptual understanding and retention in mathematics.

Realistic Mathematics Education (RME)

Rather than opening with an abstract shape definition, the lesson begins with a real, observed object. This reflects Hans Freudenthal's Realistic Mathematics Education approach, which holds that mathematics should grow out of a context that is experientially real to the student before it is formalized (Freudenthal, 1991; Gravemeijer, 1994).

Gradual Release of Responsibility

The Engage-Explore-Enrich sequence mirrors the “I do, we do, you do” structure described by Pearson and Gallagher (1983): the teacher models first (Engage), students then practice with peers (Explore), and finally receive individualized support and extension (Enrich), gradually shifting cognitive responsibility from teacher to student.

Situational Interest as an Entry Point

Opening the year with a concrete, relatable real-world object is an intentional engagement strategy. Hidi and Renninger's (2006) four-phase model of interest development identifies triggered situational interest as the necessary first step toward the kind of sustained, individual interest a teacher hopes students will eventually bring to a subject on their own.

Precision as a Mathematical Habit of Mind

The lesson's emphasis on straight lines and careful construction is not simply about neatness. It operationalizes attention to precision as a measurable, teachable habit of mind, consistent with how Mathematical Practice 6 is treated across K-12 mathematics education research and standards guidance.

What to Look For as an Observer

- Students independently deciding which geometric shapes best approximate a curved, biological form (the dragonfly) using only straight-edged manipulatives — this is MP4 and MP6 working together.
- The shift in classroom energy and talk as students move from the front board (Engage) to the back board (Explore) — notice the change from listening to negotiating ideas with peers.
- Students keeping a running tally as they build — an early, informal encounter with organizing data that will resurface formally later in the year.
- The Enrich conversation, where Professor Xavier helps individual students connect their shape tally to a “name” for their object — listen for how this personalizes an abstract skill.

Appendix: Day One in the CPA Grade 6 Math Lesson Plan Template

For administrators cross-referencing today's observation against Chandler Park Academy's required weekly lesson plan format, the table below shows how today's lesson populates that same template.

Template Field	Day One Content
Standard(s) & Taxonomy Level	MP4, MP5, MP6 C/A
Learning Target(s) (SWBAT)	SWBAT model a real-world observation using geometric shapes, use tools (graph paper, ruler, manipulatives) strategically, and construct precise geometric figures.
Success Criteria (I Can)	I can use shapes and drawings to model something I observe in the real world. I can carefully and precisely

	construct geometric figures using tools. I can represent and organize information using a tally.
Daily Marzano Strategies	Previewing New Content; Helping Students Engage in Cognitively Complex Tasks
Do-Now (5 min)	Students meet Professor Xavier Honablue, M.Ed. and settle in for Day One of Grade 6 mathematics.
Instructional Plan	Engage: whole-class observation of a real dragonfly at the front board; students simulate it with geometric shape manipulatives along a quadded, diagonal graph-paper line and build a shape tally. Explore: at the back board with classmates, students design and build their own imagined object on a second, self-quadded sheet, keeping the same tally system. Enrich: small-group time with Professor Xavier to learn how a shape tally can be used to name the simulated object.
Critical Content Debrief (5 min)	Can students accurately simulate an observed object using precise, straight-lined geometric shapes, and maintain an accurate tally of the shapes used?
Daily Formative [Ungraded] / Summative [Graded] Assessments	Formative: teacher observation during Engage and Explore, plus a tally-sheet review during Enrich. Summative: none today - completed graph-paper sheets are collected as a baseline artifact.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
Whole-class modeling at the front board before independent work; the same tools (graph paper, ruler, manipulatives) provided to every student.	Pre-drawn quadrant lines or larger manipulatives for students needing support with construction; small-group check-ins during Explore.	One-to-one support during the Enrich station; simplified tally recording (fewer shape types) as needed.

Data Source: Classroom Formative Assessments [Ungraded]; Attendance Data.

Explore More: Resources for Observers

For guests who would like to look deeper into the frameworks and standards behind today's lesson, here are curated places to start.

About Today's Object: The Dragonfly

How Math Helps Explain the Delicate Patterns of Dragonfly Wings

Science News on how researchers use mathematics to explain the geometric wing patterns in dragonflies - the same

Dragonfly Facts

A National Geographic overview of dragonfly biology and behavior, for guests curious about the real insect behind real object students simulated with shapes today.

Read on Science News

today's Engage station.

Read on Nat Geo

For Classroom Observers

NCTM: Principles to Actions

The National Council of Teachers of Mathematics' framework for effective, research-based mathematics teaching practices - useful context for what to look for during a lesson observation.

View on nctm.org

The Station Rotation Model

The Clayton Christensen Institute's overview of the station-rotation blended-learning model that the E3 Math Station System is built on.

Read the overview

For Deeper Research & Standards

Standards for Mathematical Practice

The official Common Core description of MP4, MP5, MP6, and the other five practices referenced throughout today's lesson.

View on corestandards.org

Michigan K-12 Math Standards

The Michigan Department of Education's full K-12 Standards for Mathematics document.

View Michigan DOE PDF

CASEL SEL Framework

The Collaborative for Academic, Social, and Emotional Learning's five core competencies, which ground the growth-mindset elements woven into CPA lessons.

View on casel.org

References

Freudenthal, H. (1991). Revisiting Mathematics Education: China Lectures. Kluwer Academic Publishers.

Gravemeijer, K. (1994). Developing Realistic Mathematics Education. CD-β Press.

Hidi, S., & Renninger, K. A. (2006). The four-phase model of interest development. Educational Psychologist, 41(2), 111-127.

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Pearson, P. D., & Gallagher, M. C. (1983). The instruction of reading comprehension. Contemporary Educational Psychology, 8(3), 317-344.

Witzel, B. S., Riccomini, P. J., & Schneider, E. (2008). Implementing CRA with secondary students with learning disabilities in mathematics. Intervention in School and Clinic, 43(5), 270-276.